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Technical proposal (Envelope 1)

Tender GCD/T/2026/047 · Electronic Seal (e-Seal) Tracking System for Transit Cargo

To: The Chair, Tender Committee, General Customs Directorate

Having examined the tender documents for GCD/T/2026/047, we, Trakom Security Solutions W.L.L., submit our technical proposal. This envelope contains no prices; our prices and commercial terms are in Envelope 2.

Nadia Haddad
Chief Executive Officer, Trakom Security Solutions W.L.L.

Date: 20 October 2026
Reference: TSS/GCD/047/26

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1. Company and references

Trakom Security Solutions designs and operates electronic cargo-tracking systems for customs administrations. Founded in 2012, we have 120 staff, and our TrakSeal e-seals and TrakView platform monitor more than 1.1 million transit trips a year in 4 countries.

Reference	Scope	Year
Eastern Province Customs, Al-Waha transit corridors	2,000 e-seals, 9 corridors, monitoring centre, declaration-system integration	2021
Free Zone Authority, bonded transport	600 e-seals, 4 corridors	2023
Ras Hadd Port Customs	900 e-seals, 2 border posts, CDS integration	2025

2. The TrakSeal G4 e-seal

TrakSeal G4 is a reusable cable seal with a 5 mm steel cable and a locking body of glass-fibre reinforced polymer. It is certified to ISO 17712:2013 as a high-security (H) seal by an accredited laboratory (certificate ISO17712-H-23-0418, attached as Appendix B). It is rated IP67 and operates from −20 °C to +70 °C.

- **Battery:** 45 days with one position report every 5 minutes while moving and every 60 minutes when stationary. Full recharge in 2.5 hours.
- **Positioning and communication:** GPS, GLONASS and Galileo; 4G LTE with automatic 2G fallback. Up to 25,000 events are stored when out of coverage and sent automatically on reconnection.
- **Tamper detection:** cable cut, cable removal and opening of the housing. Alerts are sent immediately; in our installations the median time to the monitoring screen is 12 seconds, and 99.9 % of alerts arrive within 45 seconds.
- **Arming and disarming:** only an officer signed in to the TrakView Officer app can arm or disarm a seal, by NFC contact with the seal. Each action is linked to the transit declaration number. Drivers have no means to arm or disarm a seal.
- **Security:** AES-256 encryption between seal and platform; every position and event is signed on the seal with its GNSS time stamp.
- **Charging:** 6 charging racks with 50 slots each, one per customs site.

3. The TrakView monitoring platform

TrakView is installed on the Directorate's virtual servers in its data centre (production) and in its disaster-recovery site; it needs 8 virtual machines (sizing in Appendix C). No data leaves the Directorate's infrastructure. Licences are perpetual and activated offline.

- Live map with the 12 corridors as geofences. Alerts for route deviation, stops longer than 30 minutes (configurable), tampering, low battery, loss of signal, and a seal opened away from its destination, with an acknowledgement workflow.
- CDS integration through REST APIs: the trip opens automatically from the transit declaration when the officer arms the seal, and closes when it is disarmed at the office of destination. Discrepancies are reported back to the CDS. We have built this integration with two declaration systems in our references.

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- Officer app for Android, in Arabic and English.
- Reports of transit times and alerts per corridor, seal utilisation and officer actions, exported to Excel and PDF.
- Officers and operators sign in with their Active Directory accounts and the Directorate's multi-factor authentication. Users and integrations connect over TLS 1.3.
- All arming, disarming and acknowledgement records are kept for 5 years.

TrakView runs active–standby across the two sites. We commit to 99.5 % availability per calendar month.

4. Implementation, training and service

Milestone	Month
Platform installed, CDS integration tested	2
Pilot: 200 seals on 2 corridors, pilot acceptance	3
Rollout to the 12 corridors	4–5
Full-rollout acceptance	6

Training. 60 officers (5 sessions of 12) and 12 monitoring-centre operators, in Arabic and English, with manuals in both languages.

Service for 3 years. 24/7 service desk in Al-Noor with a 1-hour response to platform outages. Faulty seals are replaced within 3 working days from our local stock, and a pool of 45 spare seals is kept at the Directorate's sites. SIM cards from a national operator, with data plans for all 1,500 seals for the 3 years, are included.

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5. Compliance matrix (Annex 1)

Clause	Requirement	Response	Offered
3.1	Reusable, ISO 17712 H, IP67	Comply	ISO 17712:2013 H certificate, IP67 (§2)
3.2	Battery ≥ 30 days at 5-minute reporting; recharge < 4 h	Comply	45 days; recharge 2.5 h (§2)
3.3	GNSS, 4G with 2G fallback; ≥ 10,000 events offline	Comply	25,000 events (§2)
3.4	Tamper alert within 60 s	Comply	Median 12 s, 99.9 % within 45 s (§2)
3.5	Arm/disarm by officers only	Comply	Officer app with NFC (§2)
3.6	−10 °C to +60 °C	Comply	−20 °C to +70 °C (§2)
4.1	Live map, 12 corridors, alerts	Comply	§3
4.2	CDS integration (REST)	Comply	§3
4.3	Officer app (Android), Arabic and English	Comply	§3
4.4	Reports, Excel and PDF export	Comply	§3
4.5	Hosted in the Directorate's data centre; CTS-2026	Comply	Directorate data centre and DR site (§3)
4.6	Availability ≥ 99.7 %	Partially comply	99.5 % (§3)
5.1	Pilot ≤ 3 months; full rollout ≤ 6 months	Comply	Pilot month 3, rollout month 6 (§4)
5.2	Training: 60 officers, 12 operators	Comply	§4
5.3	3-year service: 2 h response, 5-day replacement, 5 % spare pool	Comply	1 h response, 3-day replacement, 45 spare seals (§4)
5.4	Mobile data included for 3 years	Comply	National operator SIMs included (§4)

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